INCH-POUND

MIL-P-43649B(GL)
28 June 1991
SUPERSEDING
MIL-P-43649A(GL)
2 August 1972

MILITARY SPECIFICATION

PARACHUTES, CARGO, 12-, 26-, AND 34– FOOT DIAMETER

This specification is approved for use by the Natick Rsearch, Development, and Engineering Center, Department of the Army, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for three cargo parachute assemblies and the separate components thereof.

1.2 Classification.

1.2.1 Size. The cargo parachutes shall be of the following sizes as specified (see 6.2).

Size - 12-foot diamter
Size - 26-foot diameter
Size - 34-foot diamter

1.2.2 Components. The following cargo parachute components may be procured separately when specified (see6.2).

AMSC N/A

FSC 1670

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

For 12-foot diameter cargo parachute:

Canopy assembly
Deployment bag
Static line

For 26-foot diameter cargo parachute:

Canopy assambly
Deployment bag
Static line

For 34-foot diamter cargo parachute:

Canopy assembly
Pack
Drogue
Static line

2. APPLICABLE DOCUMENTS

2.1 Govemment documents.

2.1.1 Spacifications, standards, and handbooks. The following specifications, standards, and hamlbooks form a part of this document to the extent Specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

T-T-871	- Twine, Cotton, Wrappig
V-T-295	- Thread, Nylon
PPP-B-26	- Bag, Plastic (General Purpose)
PPP-B-601	- Boxes , Wed, cleated-Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-5610	Parachute Assemblies and Major Sub-Assmblies, Packaging and Packing Of
MIL-D-28024	- Drogue, Parachute Static Line

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methds
FED-STD-751	Stitches, Seams and Stitchings

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T21:56Z
Check the source to verify that this is the currentversion before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless othemise irdicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S ARMY NATRICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-216	Parachute, Cargo, Ringslot; 12 Ft. Diameter
11-1-217	Canopy Assembly; 12 Ft. Dia., Ring Slot cargo Parachute
11-1-218	Deployment Bag; 12 Ft. Diameter Cargo Parachute
11-1-219	Static Line; Cargo Prachute
11-1-507	Canopy Etails
11-1-508	Riser; Cargo Parachute
11-1-509	Panels, -
11-1-555	Prachute, Cargo, Ringslot; 26 Ft. Diameter
11-1-556	Canopy Assembly 26 Ft. Diamter Ringslot cargo Parachute
11-1-557	Canopy Details
11-1-558	Canocopy _ Sections
11-1-559	Riser; Cargo Parachute
11-1-560	Panels, Canopy
11-1–578	Parachute, Cargo, 34 Ft. Nominal Diameteer, Cotton, Bi-Conical Annular Ring, 500 Lb Capacity
11-1-579	Conopy; 34 Ft. Diameter Bi-Conical, G-14 Cargo Parachute
11-1-580	Canopy, Details and Sections
11-1-581	Riser, 34 Ft. Diameter Bi-Conical Cargo Parachute
11-1-3954	Deployment Bag Assembly; 26 Ft. HV Parachute
11-1-3955	Body Assembly
11-1–3956	Flap, Assembly
11-1-3960	Loop Assembly

Source: https://assist.dla.mil -- Downloaded: 2026t0-21T21:56Z
Check the source to verify that this is the currentversion before use.

AIR FORCE

51C6166 - Line, Static, Parachute
51D6553 - Pack, Corgo parachute
61D2459 - Drogue, Parachute Static Line

(Copies of drawings are available from the U.S. Army Natick Research, Developmennt, and Engineering Center, ATTN: STKK-UX, Natick, MA 01760-5017)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified the issues of documments not listed in the DODDISS are the issues of the documments cited in the solicitation (see 6.2) .

AMERICAN SOCIETY TESTING AND MATERIALS (ASTM)

D 3951 - Standards Practice for Cam2mial Packing

(Application for copies should be addressed to the American Society for testing and Materials, 1916 Race Street, Philadelphia, PA 19103 -1187.)

(NON-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services)

2.3 Order of procedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes procedence Nothing-inginthisd ocument, however, supersdes applicable laws and rqulations unless a specific exemption has been obtained.

3. Requirements

3.1 Front article When specified (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials. Materials shall be in accordance with the requirements specified by the applicable drawings (see 2.1.2) , subsidiary documents applicable thereto, and as specified herein. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification

3.3 Construction and assembly. The items (see 6.2) shall conform to the requirements of this specification and to design, details, and dimensions shown or specified on the applicable drawings.

3.3.1 Stitching. All stitching shall be in accordance with FEDS-STD-751. Broken threads, skipped and missing stitches occurred in type 301 or 401 stitching shall be overstitched not less than 1 inch past ends of the defect. Broken threads and skipped stitches occurring in type 308 stitching shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

over-stitched not less than three stitches past the ends of the defect. Except where atherwise specified, all type 301 or 401 stitching shall be back-stitched not less than 1/2 inch, unless ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so that loose stitching does not result and the lock will be imbedded in the materials sewed. Thread ends shall be trimmed throughout. Runoffs shall be restitched to conform to the specified stitch pattern.

3.3.1.1 Stitching, automatic machinery Stitching patterns produced by autom-matic machines may be utilized in lieu of those specified, providing seam strength efficiency and fuction equal to, or better than, the specified pattern, and upon approval by the procuring activity. Certification of equivalency, supported by enineering data, may be required.

3.3.1.2 Darns. Holes, cuts, or tears in the canopy, deployment bag or pack cloths shall be darned with thread, type I or II, class 1, size A, color OD S-1 of V-T-295. Damaged areas of the cloth which can be darned shall not exceed 1/4 inch in diamter for holes (except drill holes) and 1/2 inch in length for cuts or tears in any direction. Fourteen daerns are allowed per conopy, four per deployment bag, and four per pack.

3.3.2 Canopy permeability. The air permeability of the cloth of the com-pleted canopy shall not exceed the specified liner limit by more than 10 percent nor the Specified upper limit by more than 10 pement when tested in accordance with 4.4.5.

3.3.3 Canopy Patching. One sewn patch is allowed per canopy. The maximum damaged area that can be patched is an area by 7 inches. the. patch shall be located no closer than 1 inch from a radial seam, edge reinforcment, or lower lateral band. The patch shall be square or rectangular in shape and conform to the patching proceduring in 3.3.4.

3.3.4 Patching procedure.

a. Plain the repairable item on a repair table, smooth the fabric around the damaged area, and secure the item to the table with pushpins. Do not pin the damaged area.

b. Using an authorized marking aid of constrasting color, mark a square or rectangle around the area to be patched and insure that one side of the marked square or rectangle is parallel to the warp or filling of the fabric.

c. Cut the damaged area fabric along the lines made in (b) above. Further cut the fabric diagonally at each corner to allow a 1/2 inch fold back in the raw edges.

d. Make a 1/2 inch foldback on each raw edge.Pin and baste each fold-back to complete the prepared hole.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

e. Using the same type material as in the original instruction, mark and cut a patch 2-1/2 inches wider and longer than the inside measurements of the prepared hole. Insure that the patch material is marked and cut along the warp or filling of the fabric.

f. Center the patch material over the prepared hole and insure the warp or filling of the patch material matches the warp or filling of the fabric being patched. Pin the patched material in position.

g. Make a 1/2 inch fold under on each edge of the patch material and baste the patch to the prepared area.

h. Remove the pushpins securing the item to the repair table and secure the patch by machine stitching, using the applicable details in figure 1. The stitching should be accomplished with thread that is contrasting in color to the fabric. Thread shall be V-T-295, size “E” nylon, type I or II, class A, type 301, using 7 to 11 stities per inch in aocordance with FED-STD-751. Back stitch 1/2 inch minimum or lock by at least 2 inches minimum and not more than 4 inches at each end of a stitch row.

3.4 Workmanship. The workmanship shall conform to the quality of product establish by this specification and the of defects shall not ex- the applicable acceptable quality levels.

4. Quality ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations ard tests) as spcified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to prform any of the inspections set forth in this specification where such inspections are deemed necessary to ensured supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 of 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submittal to the Covernmnt for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all Specified dimension have been met. When dimensions cannot bens

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requimments

4.2 Classification of inspectios The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be xamined for the defects specified in 4.4.3 and 4.4.4 and exammined shall be tested for the characteristic Specified in 4.4.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requists of referenced documnts unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-Process inspection of subasmblies shall be made to ascertain that instruction details which cannot examination in the finished product are in accordance with Specified requirements. The reserves Gov the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.3 End item visual examination.

4.4.3.1 Canopy with riser. The canopies shall be examined for the applicable defects listed in table I. The lot size shall be in units of canopies and risers. The sample unit shall be one canopy and riser. The inspection level shall be S-2 and the acceptable quality level (AQL) , expressed in terms of defects per hundred units, shall be 1.0 for major defects and 6.5 for total (major and minor combind) defects.

TABLE I. Canopy visual defects

Defect	Classification Major	Classification Minor
cloth, webbing and tape		
Any hole in cloth (except drill holes)		
1/4 inch diameter or less		201
More than 1/4 inch diameter	101	
color not as specified		202
Not specified type or class	102	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE I Canopy visual defects (cont'd)

Defect	Classification Major	Minor
Cloth, webbing and tape (cont'd)		
Cut or tear in cloth -		203
1/2 inch or less	103	
More than 1/2 inch	104	
Multiple floats	105	
Grease or oil stain	106	
Embedded crease or wrinkle in excess of 1/2 inch width	107	
	108	
Patches		
More than one per canopy	109	
Patch size exceed- 7 inches by 7 inches	110	
Location not as specified	111	
Shape not as Specified	112	
Procedure, not as specified in 3.3.3	113	
Cord and thread		
Not specified type or size as applicable	114	
Color not as specified		204
vent reinforement overlap		
vent reinforcement overlap does not span two main seams	115	
vent reinforcment band, improper assembly	116	
Vent line length (across vent) not specified	117	
Less than 1/4 inch		205
1/4 inch or more		
Twisted vent line		206
More than 360°	118	
More than 360°, causing kinks in line		
Gore width at vent, measurement C/L to C/L on adjacent radial seam, not as specified		207
Less than 1/4inch	119	
1/4 inch or more		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE I Canopy visual defects (cont'd)

Defect	Classification Major	Minor
Slot width, not as specified		
Less than 1/8 inch		208
1/8 inch or more	120	
skirt reinforcementt band improperly assembled	121	
Underfold (raw edge) skirt hem	122	
NOTE: When a raw edge is the result of overfolding the edge extending more than 1/2inch but less than 1 inch, the raw edge may be trimmed and will not be scored as a defect. If the condition extends more than 1 inch from the inside edge of the skirt hem, the stitching will be removed and the hem refolded and restitched.		
Tucks, wherein section of gore fabric is caught in stitching of hem	123	
Suspension lines crossed	124	
Cut, frayed or damaged, suspension or vent lines	125	
Suspention line 1-, not as specified		
Less than 3/4 inch		209
3/4 inch or more	126	
Suspension line twisted		
More than 360°		210
More than 360°, causing kinks	127	
Stitches and stitching		
wrong type	128	
TWO or more stitches per inch less than specified	129	
TWO or more stitches per inch more than specified		211
Broken, skipped or missing stitches not overstitched as specified	130	
Open seam	131	
NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seam will not be scord as defects.		

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:56Z
Check the source to verify that this is the currentversion before use.

TABLE I. Canopy visual defects (cont'd)

Defect	Classification Major	Minor
Stitches and stitching (con"t)		
Wrong type seam	132	
Missing incomplete stitch pattern	133	
Runoff not restitched in accordance with specified pattern	134	212
Zig-zag Stitching not sewn through suspension lines		
Tension loose, result-in loose top or bobbin thread; or tight tension, resulting in tightness or puckering of material		213
		214
Thread ends not trimmed		215
Ends of stitching not secured as specified		
Stitch pattern length-		216
Less than 1/8 inch minus	135	
1/8 or more minus		217
1/2 inch or more plus	136	
Stitch pattern not cmplete	137	
Needle or feed dog damge		
Length of components and location dimension not othewise covered, not as specified		
up to 5 inches		218
Less than 1/8 inch	138	
1/8 inch or more		
5 inch or over -		219
Less 1/4 inch	139	
1/4 inch or more		
Riser		
Length overall, not as specified by -		220
Less than 1/2 inch	140	
1/2 inch or more		
Loop length, not as specified by -		221
Less than 1/2 inch	141	
1/8 inch or more		
Insertion of suspension line in core at skirt attachment loop and riser attachment point not as specified by -		222
Less than 1/4 inch	142	
1/4 inch to 1/2 inch		223
More than 1/2 inch plus		224

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE I. Canopy visual defects (cont'd)

Defect	Classification Major	Minor
Suspension line attachment loop length at skirt bad not as specified by -		
less than 1/8 inch		225
1/8 inch or more	143	
Any component missing	144	
ANY incomplete assembly	145	
Identification marking illegible, incomplete, or missing		226
Clevis		
Component missing	146	
Part fractured, malformed, or damaged	147	
Corrosion or rust	148	

4.4.3.2 Deployment bag. The bags shall be examined for the applicable defects listed in table II. The lot size shall be expressed in units of deployment bags. l%e sample unit shall be one deployment bag. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. Deployment bag visual defects

Defect	Classification Major	Minor
cloth, webbing and tape		
Any hole in cloth (except drill holes) -		
1/4 inch diameter or less		201
More than 1/4 inch diameter	101	
Color not as specified		202
Not specified type or class	102	
Cut or tear in cloth		
1/2 inch or less		203
More than 1/2 inch	103	
Darns, over 5 per bag	104	
Thread		
Not specified type or size	105	
Color not as Specified		204

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T21:56Z
Check the source to verify that this is the currentversion before use.

TABLE II. Deployment bag visual defects (cont'd)

Defect	Classification Major	Minor
Finished dimensions, body and flap (prior to assembly)		
Less than 1/4 inch less than specified	106	205
1/4 inch or more than specified		206
More than 1/2 inch more than specified		
Stitches and stitchings		
Wrong type	108	
Two or more stitches per inch than specified		207
Two or more stitches than specified		
Broken, skipped, or missing stitches overstitched	109	
as specified	110	
Open seam		
Note: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seams will not be scored as defects.		
Runoff not restitched in accordance with specified pattern		208
Tension loose, is resulting in exposed top or bobbin thread, or tight tension, resulting in tightness or puckering of material		209
		210
Ends of stitching not secured as specified		
		211
	111	
		212
	112	
	113	
Needle or feed dog damage		
Length of components and location dimensions, not otherwise covered, not as specified		
Up to 5 inches		213
Less than 1/8 inch	114	
1/8 inch or more		
5 inch or more		214
Less than 1/4 inch	115	
1/4 inch or more		
	116	
Partial assembly or any comonents missing		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

Table II. Deployment bag visual defects (cont'd)

Defect	Classification Major	Minor
Misassembly	117	
Marking illegible, incorrect or missing		215

4.4.3.3 Pack. The pack shall be examine for the applicable defects listed in table III. The lot size shall be expressed in units of packs. The sample unit shall be one pack. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 10 for total (major and minor combined) defects.

TABLE III. Pack visual defects

Defect	Classification Major	or Minor
cloth, webbing and tape		
Any hole in cloth (except drill holes) –		
1/4 inch diameter or less		201
More than 1/4 and diameter	101	
Color not as specified		202
Not specified type or class	102	
Cut or tear in cloth -		
1/2 inch or less		203
More than 1/2 inch	103	
Darns, over 5 per pack	104	
Thread		
Not specified type or size	105	
Color not as specified		204
Finished dimensions, Panel assemblies (prior to assembly)		
Less than 1/4 inch less than specified		205
1/4 inch or more less than specified	106	
More than 1/2 inch more than specified		206
Stitches and stitchings		
Wrong type	107	
TWO or more stitches per inch less than Specified		207
TWO or more stitches per inch more than specified	108	
Broken, skipped, or missing stitches not overstitched as specified	109	
Open seam	110	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE III. Pack visual defects (cont'd)

Defect	Classification Major	Minor
Stitches and stitchings (cont'd)		
NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seams will not be scored as defects.		
Runoff not restitched in accordance with specified pattern		208
Tension loose, resulting in exposed top or bobbin thread, or tight tension, resulting in tightness or puckering of material		209
Ends of stitching not secured as specified		210
Stitch pattern lengths not as specified by –		
Less than 1/8 inch		211
1/8 inch or more	111	
Thread ends not trimmed		212
Missing or incomplete stitch pattern	112	
Needle or feed dog damge	113	
Length of components and location dimensions, not otherwise coverd, not as specified		
Up to 5 inches -		
Less than 1/8 inch		213
1/8 inch or more	114	
5 inches or more -		
Less than 1/4 inch		214
1/4 inch or more	115	
Partial assembly or any component missing	116	
Disassembly	117	
Marking illegible, incorrect, or missing		215

4.4.3.4 Static line. The static lines shall be examird for the applicable defects listed in table IV. The lot size expressed in the units of static lines. The sample unit shall be one static line. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 10 for total (major and minor combined) defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Static line visual defects

Defect	Classification Major	Minor
Webbing		
Any hole, cut, tear, or abrasion	101	
Not specified type or class		201
Color not as Specified	102	
Length of static line not as specified by -		
Up to 1/2 inch		202
1/2 inch or more	103	
Loop length		
Up to 1/4 inch less than specified		203
1/4 inch or more less than Specified	104	
More than 1/4 inch more than Specified		204
Component missing	105	
Stitching		
Wrong type	106	
Missing or incomplete stitch pattern	107	
Runoff not restitched in accordance with Specified stitch pattern		205
Tension loose, resulting in loose top or bobbin thread, or tight tension, resulting in tightness or puckering of material		206
Ends of stitching not secured as specified		207
Broken, skipped, or missing stitches not overstitched as specified	108	
Open seam	109	
NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or where two or more skipped stitches occur. Repairs of open seam will not be scored as defects.		
Stitch pattern lengths not as specified by -		
Less than 1/8 inch		208
1/8 to 1/4 inch	110	
1/4 inch or more plus		209
Thread ends not trimmed		210
Ends of webbing not seared		211
Thread:		
Not specified type or size	111	
Color not as specified		212

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE IV. Static line visual defects (cont'd)

Defect	Classification Major	Minor
Identification marking illegible, missing, or incorrect		213
Clevis		
Component missing	112	
Any part fracturd, malformed, or damged	113	

4.4.3.5 Drogue. The drogue shall be examined for the applicable defects listed in table V. The lot size shall be expressed in units of drogues. The sample unit shall be one drogue. The inspection level shall be S-1 and AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 10 for total (major and minor Combined) defects.

TABLE V. Drogue visual defects

Examine	Defect	Classification Major	Minor
Material	Wrong color		201
	Not specified type or class		202
Design	Any characteristic not in accordance with specified requirement	101	
Construction and assembly	Any component missing or not assembled or joined as Specified	102	
cleanness	Grease, dirt, oil, or other foreign matter		203
Seams	Open seam	103	
	Open seam averstitchd less than 1 inch		204
	NOTE: A seam shall be classified as open whenever breaks, runoffs, or two or more connsecutive skips occur in stitching and are not overstitched or restitched as specified.		
Stitches and stitching	Not type of stitch Specified	104	
	Ends of stitching not secured as specified (except where turned under in a hem or held down by other stitching)		205

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE V. Drogue visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Stitches and stitching (cont'd)	One stitch per inch less than minimum specified		206
	Two or more stitches per inch less than minimum specified	105	
	One or stitiches per inch in excess of maximum Specified		207
	Loose tension, resulting in loose top or bobbin thread; or tight tension resulting in tightness or puckering of material at stitch line		208
	Thread ends not trimmed		209
	Any row of stitching missng	106	
	Needle chews	107	
	Stitching pattern not cmplete	108	
	Stitch margin or gage not as specified		210
	Stitching run-off at edge cutting material	109	
Nylon cloth and tape	Holes, tears, or cuts	110	
	Abrasion or thin material	111	
Nylon cord	Cut, tear, or break	112	
	Abrasion or seared area (except on ends of cord)	113	
	Any Splice or not continuous		211
	Ends not seared		212
Identification marking	Missing, illegible, incorrect, wrong color or location, or not applied as Specified		213

4.4.4 End item dimensional examination. The end items (complete parachute assemblies or separate compments, as applicable) shall be examined for conformance to the dimensions specified on the drawings. Only those dimension that can be evaluated without damging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of end item. The sample unit shall be one end item (see table VI) .

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

TABLE VI. Classification of lot size, sample unit, inspection level and AOL for end item dimensional examinations.

Component	Lot size expressed in units of	Sample unit	Inspection level	AQL
Canopy with riser	Canopies with risers	1 Canopy with riser	S-1	2.5
Deployment bag	Deployment bags	1 Deployment bag	S-1	2.5
Pack	Packs	1 Pack	S-1	2.5
Static line	Static lines	1 Static line	S-1	2.5
Drogue	Drogues	1 Drogue	S-1	2.5

4.4.5 Ends of Items Testing. One completely fabricated canopy shall be tested as specified in 4.5.1 for conformnce to the air permability requirements of 3.3.2. The lot size shall be the number of canopies produced in one week. The sample size shall be the number of end item indicated by inspection level S-2. Any test failure shall because for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per units, shall be 2.5.

	Defect
Marking	Ommitted; incorrect: illegible; Of improper size, location, squence or method of application
Materials	Any component missing, damaged, or not as Specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Nmber per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items Shall be exmined for the defects listed below. The lot size shall bee expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully Packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

Examine	Defect
Finished dimensions	Length, width, or height exceed specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceed maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection

4.5.1 Canopy permeability test. The canopy permeability shall be determined in accordance with Method 5450 of FED-STD-191, except that three permeability readings shall be taken at widely separated points in each of the alternate gores around the canopy, and the average of the results determined.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as Specified (see 6.2).

5.1.1 Level A preservation

5.1.1.1 Cargo parachute, complete assembly. Each complete cargo parachute shall be unit packed in accordance with the applicable level A requirments for type II parachutes of MIL-P-5610.

5.1.1.2 Components , when procured separately (see 6.2)

5.1.1.2.1 Canopy Assembly Each canopy assembly shall be unit packed in accordance with the applicable level A requirements for type 11 parachutes of MIL-P-5610.

5.1.1.2.2 Deployment bag. Each deployment bag shall be compactly folded and then unit packed in a close-fitting 0.003-inch nominal thickness polyethylene bag conforming to type I or II, style 1 of PPP-B-26. Prior to or during the closure cperation, excess air within the bag shall be expend.

5.1.1.2.3 Static line. Each static line shall be compactly folded and securely tied with twine conforming to type I or II, 8-ply of T-T-871. Each static line shall then be unit packed in a close-fitting polyethylene bag as specified in 5.1.1.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

5.1.1.2.4 Pack (size 34-foot dia.) Each pack shall be folded to measure approximately 17 by 7-3/4 by 2-1/2 inches. Each folded pack shall be unit packed in a plastic bag conforming to type I or II, style 1 of PPP-B-26 with a nominal thickness of 0.003 inch. Prior to or during the closure operation, excess air within bag shall be expelled.

5.1.1.2.5 Drogue (Size 34-foot dia). Each drogue shall be unit packed in accordance with the applicable level A requirements of MIL-STD-38024.

5.1.2 Commericial preservation. Each complete cargo parachute and component, when procured separately (see 6.2), shall be preserved in accordance with ASTM D 3951.

5.2 Pacing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing

5.2.1.1 Cargo parachutes, complete assemblies. Eight 12-foot diameter, three 26-foot diameter, or two 34-foot diameter cargo parachutes, preserved as specified in 5.1, shall be packed in accordance with the applicable level A requirements mments for type II parachutes of MIL-P-5610.

5.2.1.2 Components, when procured separately (see 6.2)

5.2.1.2.1 Canopy assemblies. Ten 12-foot diameter, three 26-foot diameter, *or two* 34-foot diameter canopy assemblies, preserved *as specified in 5.1, shall* be packed in accordance with the applicable level A requirements for type II. parachutes of MIL-P-5610.

5.2.1.2.2 Deployment bags. Twenty-four deployment bags, preserved as specified in 5.1, shall be packed on end four in length, ten in width in a shipping container conforming to style RX-L, grade V2s of PPP-B-636, or overseas types style A or I, grade A or B, type 2 load of P?PP-B-601. Each box shall be closed, waterproofed, and reinforced in accordanoe with the appendix of the applicable documents Fiberbid. boxes shall be arranged in unit loads in accordance for the type and class of load Specified. Strapped shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.1.2.3 Static line. One hundred static lines, preserved as specified in 5.1, shall be packed as Specified in 5.2.1.2.2.

5.2.1.2.4 Pack (size 34-foot dial. Forty packs) preserved as specified in 5.1, shall be packed as specified in 5.2.1.2.2.

5.2.1.2.5 Drogue (size 34-foot dia.). Drogue shall be packed in accordance with the applicable level A requirements of MIL-STD-38024.

5.2.2 Level B packing.

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:56Z
Check the source to verify that this is the current version before use.

5.2.2.1 Cargo parachutes, complete assemblies. Eight 12-foot diameter, three 26-foot diameter, or two 34-foot diameter cargo parachutes, preserved as specified in 5.1, shall be packed in accordance with the applicable level B requirements for type II parachutes of MIL-P-5610.

5.2.2.2 Components, when procured separately (see 6.2.1)

5.2.2.2.1 Canopy assemblies. Ten 12-foot diameter, three 26-foot diameter or two 34-foot diameter canopy assemblies, preserved as specified in 5.1, shall be packed in accordance with the applicable level B requirements for type II. parchutes of MIL-P-5610 .

5.2.2.2.2 Enployment bags. Twenty-four deployment bags, preserved as specifi-fied in 5.1, shall be packed in a snug-fitting shipping container conforming to style RX-L, type CF (variety SW) or SF, class damestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method 11 as specifie in the appendix of PPP-B-636.

5.2.2.2.3 Static line. One hundred static lines, **preserved** as specified in 5.1, shall be packed as specified in 5.2.2.2.2.

5.2.2.2.4 Pack (size 34-foot dia.). Forty packs, preserved as specified in 5.1, shall be packed as specified in 5.2.2.2.2.

5.2.2.2.5 Drogue (size 34-foot dia). Drogue shall be packed in accordance with the applicable level B requirements of MIL-D-38024.

5.2.2.2.6 Weather resistant shipping container. When specified (see 6.2) the shipping container specified in 5.2.2.2.2 shall be a grade V3c, V3s, or V4s fiberboard box fabericated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

5.2.3 Commerical packing Complete cargo parachutes and camponents, when procured separately (see 6.2), presemdas specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), complete cargo parachutes and sepratelY Procured components packed as specified in 5.2, shall be palletized on a 4-way pallet in accordance with load type I or Ia of MIL-STD-147. Pallet types shall be type I (4-way-entry), type IV accordance with MIL-M-147. Pallets shall be fabricated from groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with strops in accordance with bonding means C and D or film bonding F or G. Pallet patterns shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase, unit packs, shipping containers, and palletized unit loads shall be marked accordance with the applicable marking requirements of MIL-P-5610 and MIL-STD-129 or ASTM D 3951, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The 12-foot and 26-foot diameter parachutes are intended for use in air dropping non-fragile cargo at high rates of descent. The 34-foot diameter parachute is intended for use in airdropping cargo at normal rates of descent.

6.2 Acquisition requirements Acquistion documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2.1).
c. Parachute component(s), when procured separately (see 1.2.2, 5.1.1.2, 5.2.1.2, and 5.2.2.2).
d. Issue of EODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
e. men a first article is rquired (see 3.1, 4.3, and 6.3).
f. Kvels of preservation and packing (see 5.1 and 5.2).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.2.6).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instruction in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Airdrop	Deployment bag
canopy	Pack
Drogue	Static line

6.5 Changes from Previous issue. Margiml notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026t0-21T21:56Z
Check the source to verify that this is the currentversion before use.

Custodian:

Army-GL

Review activity:

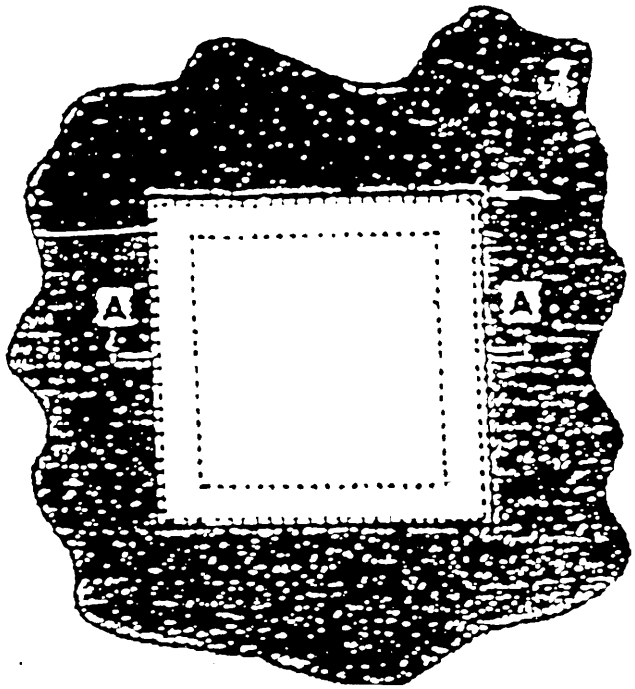
Army-AV

Preparing activity:

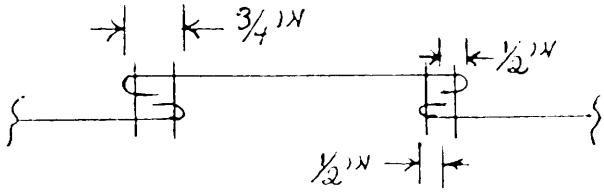
Army-GL

(Project 1670-A781)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.



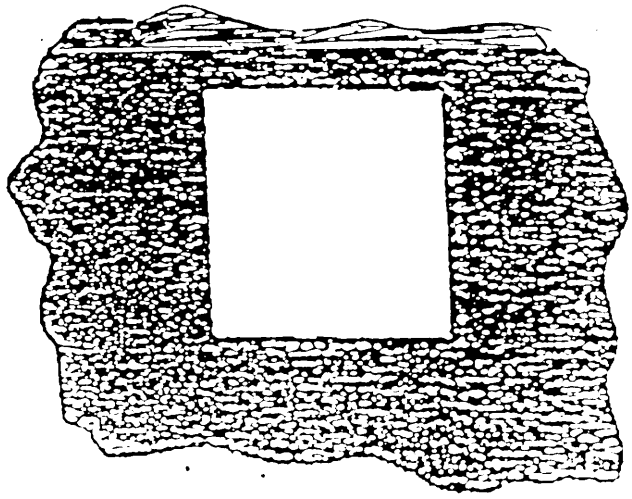
INSIDE VIEW



SECTION A-A

ALL STITCH
1/8 INCH FI

TOLORENCE:-



OUTSIDE VIEW

FIGURE 1. Sewn patch application to solid parachute ca

Source: https://assist.dla.mil -- Downloaded: 202610-21T21:56Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-43649B(GL)	1991 June 28

3. DOCUMENT TITLE
PARACHUTES, CARGO, 12-, 26-, AND 34- FOOT DIAMETER

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP3, JUL 91 *Previous editions are obsolete.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:56Z
Check the source to verify that this is the current version before use.